

Computer Organisation: physical aspects of computer systems. (comp. engineering)
(e.g. circuit design) → How does a computer work?

Computer Architecture: logical aspects of system implementation (comp. science, programming)
(e.g. instruction sets, data types) → How do I design a computer?

Conditions to be a Computer System:

- **Processor** → interprets and executes programs
- **Memory** → stores data and/or programs
- **Mechanism** → transferring data to the outside world (input-output)

Computer Level Hierarchy

6. **User** → applications that users interact with, such as Firefox
5. **High-Level Languages** → programming languages such as Java, first compiled and interpreted, then assembled before execution
4. **Assembly Language** → translates high-level program instructions, assembly instructions translate one-to-one into machine level instructions, more detailed steps
3. **System Software** → Multiple controls executing processes on the system, protects system resources, complex systems called Operating Systems, ensures that programs get fair access to the computer hardware (resources), assembly instructions must pass through
2. **Machine** → (Instruction Set Architecture Level), instructions particular to this architecture of a machine, need no compilers/interpreters/assemblers, executed immediately
1. **Control** → control unit (processor) of the computer decodes and executes instructions, can be microprogrammed (written in low level language implemented by the hardware) or hardwired (hardware directly executes machine instructions)
0. **Digital Logic** → digital circuits (gates and wires) and microchips, implement the mathematical logic of all other levels

How to Measure a Computer?

Processor Speed $\rightarrow$ frequency measured in Hertz (Hz)

Memory $\rightarrow$ size measured in bytes (B)

SI Prefixes

Prefix	Symbol	Value
tera-	T	10^{12}
giga-	G	10^9
mega-	M	10^6
kilo-	k	10^3
milli-	m	10^{-3}
micro-	μ	10^{-6}
nano-	n	10^{-9}
pico-	p	10^{-12}

"i" $\rightarrow$ powers of 2

However, it is sometimes more accurate to measure memory in powers of 2:

Prefix	Symbol	Value
tebi-	Ti	$2^{10 \times 4}$
gibi-	Gi	$2^{10 \times 3}$
mebi-	Mi	$2^{10 \times 2}$
kibi-	Ki	$2^{10 \times 1}$

Von Neumann Model